

Causal Effect of Payment Protection Plans on Credit Default Risk

MATH-516

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Abstract

This paper investigates the causal effect of enrolling in a Payment Protection Plan (PPP) on the probability of credit card default. Using the potential-outcomes framework, we estimate the Average Causal Effect (ACE) on the risk-difference scale from observational data. To address self-selection and the practical challenge of choosing an optimal adjustment set—balancing the variance inflation of over-adjustment against the residual confounding of under-adjustment—we compare Outcome Regression (g-computation) and Inverse Probability Weighting (IPW). We evaluate four adjustment strategies: full-covariate adjustment, outcome-focused L1-penalization (LASSO), a global Rank-PC screening procedure, and a local IAMB Markov-blanket search.

For the graphical discovery step, we adapt the rank-PC construction (Harris and Drton, 2013 [1]) to robustly estimate latent correlations, aligning with nonparanormal theory. Empirically, the estimated ACE remains robustly positive across all fitted models, suggesting that PPP enrollment is associated with higher default risk. Overall, the results support a positive causal effect and demonstrate that, in this empirical context, discovery-based reduction should be interpreted as a screening device rather than a definitive recovery of the underlying causal DAG.

1 Introduction

In the context of retail banking and credit risk management, Payment Protection Plans (PPP) are financial products designed to cover credit card minimum payments under adverse life events. However, evaluating the true efficacy of such programs poses a significant statistical challenge. A naive comparison between customers who enroll in a PPP and those who do not is inherently flawed due to self-selection bias: the decision to enroll is heavily influenced by underlying financial behaviors, demographic factors, and baseline credit risk. In causal inference terms, the observed association is entangled with confounding bias, rendering correlation strictly unequal to causation.

Let $\mathcal{D} = \{(A_i, Y_i, L_i)\}_{i=1}^N$ be our observational dataset drawn from the population, where for each individual i , $A_i \in \{0, 1\}$ represents the binary treatment assignment (1 if enrolled in the PPP, 0 otherwise), $Y_i \in \{0, 1\}$ represents the binary outcome (1 if the customer defaults on the next payment, 0 otherwise), and $L_i \in \mathbb{R}^p$ is a p -dimensional vector of observed pre-treatment covariates.

The naive statistical estimand is the crude risk difference

$$\Delta_{\text{naive}} = \mathbb{E}[Y \mid A = 1] - \mathbb{E}[Y \mid A = 0].$$

Because the covariates in L may affect both treatment selection and outcome risk, Δ_{naive} does not isolate the causal effect of treatment. To formalize the target quantity, we adopt the potential-

outcomes framework [2] and define the Average Causal Effect (ACE) as

$$\text{ACE} = \mathbb{E}[Y^{a=1}] - \mathbb{E}[Y^{a=0}],$$

where Y^a denotes the outcome that would have been observed had the individual, possibly contrary to fact, received treatment level a .

The main objective of this project is to bridge the gap between the observational distribution $P(Y, A, L)$ and the potential outcomes under intervention. In practical terms, this requires identifying an adjustment set $L^* \subseteq L$ that is rich enough to control confounding but not so large as to destabilize estimation. In high-dimensional settings, full adjustment may inflate variance and worsen positivity, whereas overly aggressive reduction may leave residual confounding.

To address this, we compare four adjustment strategies within a single inferential framework. First, we use the full observed covariate vector L as a benchmark adjustment set. Second, we consider a reduced set obtained through ℓ_1 -penalized logistic regression focused on the outcome Y . Third, we use a global causal-discovery approach based on a rank-PC procedure. Following Harris and Drton (2013) [1], this step estimates a latent correlation matrix from rank correlations and then computes partial correlations from the appropriate principal submatrices before testing conditional independences. Fourth, we complement this with a local causal-discovery approach using the Incremental Association Markov Blanket (IAMB) algorithm, which directly targets the Markov blanket of the outcome and therefore provides a compact candidate adjustment set.

For each resulting adjustment set, the ACE is then estimated using two standard estimators: Outcome Regression, implemented through g-computation, and Inverse Probability Weighting (IPW). By comparing these estimators across adjustment strategies, and by examining associated diagnostics such as propensity-score overlap and standardized mean differences, the analysis evaluates whether the Payment Protection Plan appears to have a protective causal effect on default risk or instead acts as a marker of elevated risk.

2 Data Architecture and Descriptive Evidence of Confounding

2.1 Characterization of the Covariate Space

The empirical analysis is conducted on the dataset, comprising an independent and identically distributed (i.i.d.) sample of $N = 500$ credit card clients. Formally, the observed data can be denoted as a collection of random vectors $\mathcal{D} = \{(A_i, Y_i, \mathbf{L}_i)\}_{i=1}^N$. Within this collection, $Y_i \in \{0, 1\}$ serves as the binary outcome indicator, taking the value of 1 if the client defaults on the subsequent month’s payment and 0 otherwise. The variable $A_i \in \{0, 1\}$ represents the binary treatment indicator, where 1 denotes enrollment in the Payment Protection Plan and 0 indicates non-enrollment. Finally, $\mathbf{L}_i \in \mathbb{R}^p$ is a p -dimensional vector of pre-treatment covariates ($p = 22$).

The covariate space is heterogeneous and clearly departs from multivariate Gaussianity. It contains continuous monetary variables such as LIMIT_BAL, BILL_AMT1–BILL_AMT6, and PAY_AMT1–PAY_AMT6. Furthermore, it includes ordinal variables such as EDUCATION and PAY_0–PAY_6, alongside binary or nominal variables like SEX and MARRIAGE. In particular, the repayment-status variables are ordered measures of payment delay, a structural property that naturally motivates the use of rank-based methods in the discovery phase.

2.2 Baseline Imbalance and Selection Bias

Before applying causal adjustment techniques, it is useful to assess whether treatment assignment appears close to random in the observed data. If PPP enrollment were randomized, one would

expect $A \perp\!\!\!\perp \mathbf{L}$. Instead, the dataset exhibits clear systematic differences between treated and untreated groups.

A naive estimate of the treatment effect gives the crude risk difference

$$\hat{\Delta}_{\text{crude}} = \hat{\mathbb{E}}[Y \mid A = 1] - \hat{\mathbb{E}}[Y \mid A = 0] \approx 0.116,$$

which suggests that PPP enrollment is associated with an 11.6% increase in default probability. To evaluate whether this crude difference may be confounded, we compare baseline covariate distributions across treatment groups using standardized mean differences (SMDs),

$$\text{SMD}_j = \frac{|\bar{x}_{1,j} - \bar{x}_{0,j}|}{\sqrt{(s_{1,j}^2 + s_{0,j}^2)/2}},$$

where $\bar{x}_{a,j}$ and $s_{a,j}^2$ are the sample mean and variance of covariate L_j in treatment group $a \in \{0, 1\}$.

As shown in Figure 1(b), the strongest pre-treatment imbalances occur for **AGE**, **EDUCATION**, and **LIMIT_BAL**, with absolute SMDs approximately 0.867, 0.410, and 0.305, respectively. Thus, the crude comparison between PPP enrollees and non-enrollees is unlikely to have a causal interpretation without adjustment.

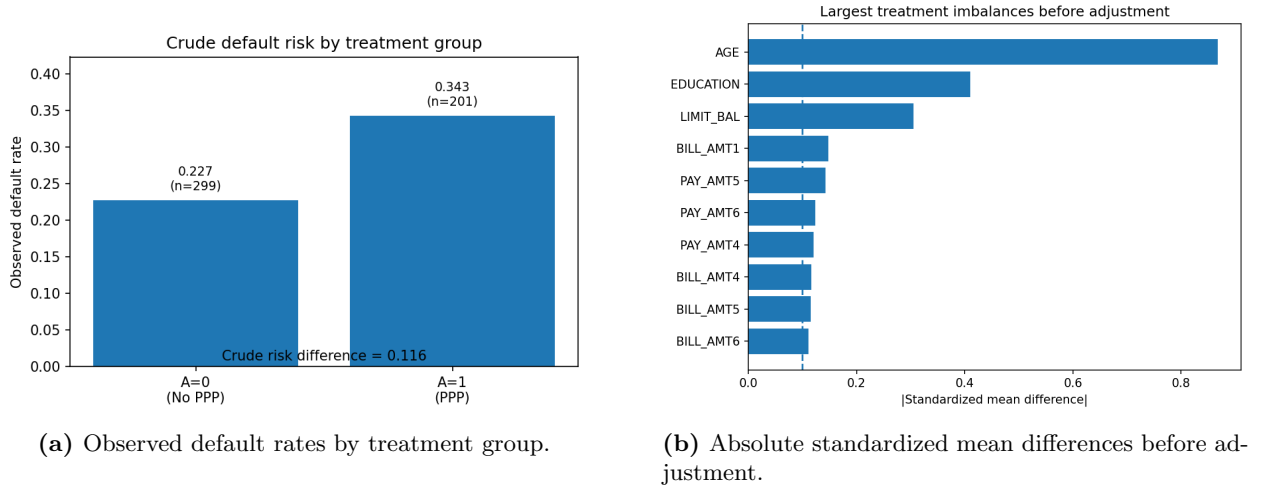


Figure 1

3 Theoretical Framework and Assumptions

3.1 The Potential Outcomes Framework

To evaluate the causal effect of the Payment Protection Plan (PPP) on credit default, we adopt the potential-outcomes framework. Let $A \in \{0, 1\}$ denote treatment and $Y \in \{0, 1\}$ denote outcome. For each individual i , define two potential outcomes: $Y_i^{a=1}$, the outcome if enrolled in the PPP, and $Y_i^{a=0}$, the outcome if not enrolled.

The causal estimand of interest is the Average Causal Effect (ACE) on the risk-difference scale:

$$\text{ACE} = \mathbb{E}[Y^{a=1}] - \mathbb{E}[Y^{a=0}].$$

Identification of the ACE from observational data relies on three standard assumptions:

1. **Consistency:** if an individual actually receives treatment level $A = a$, then the observed outcome equals the corresponding potential outcome, $Y = Y^a$.
2. **Positivity:** every covariate pattern with positive density has nonzero probability of receiving either treatment level,

$$0 < \Pr(A = 1 \mid L = l) < 1 \quad \forall l \text{ such that } \Pr(L = l) > 0.$$

3. **Conditional exchangeability:** the potential outcomes are independent of treatment conditional on a sufficient adjustment set L ,

$$Y^a \perp\!\!\!\perp A \mid L.$$

The central practical challenge is therefore to choose adjustment sets that make these assumptions as plausible as possible while keeping estimation stable.

4 Methods and Mathematical Formulations

4.1 Rank-Based Latent Correlation Estimation

A major obstacle in the present dataset is that several covariates depart substantially from the setting in which standard Gaussian graphical procedures are most naturally justified. In particular, variables such as `EDUCATION` and the repayment-status measures `PAY_0` to `PAY_6` are ordinal, the treatment and outcome variables are binary, and many financial variables are strongly skewed and heavy-tailed. If one were to apply Gaussian conditional-independence testing directly to the raw data, the interpretation of Pearson partial correlations would be questionable, and the resulting graph-learning step could become highly sensitive to the marginal scale of the variables.

To address this issue, the graphical-discovery stage is built on the rank-PC construction of Harris and Drton (2013) [1]. The underlying idea is to assume that the observed variables are monotonic transformations of an unobserved latent Gaussian vector. Formally, this relies on a nonparanormal (or Gaussian copula) framework, which states that while the observed continuous data may be highly skewed, there exists a hidden, perfectly normally distributed set of variables driving the true dependence structure.

Strictly speaking, the nonparanormal framework assumes that the observed variables are continuous and that the transformations linking the latent variables to the observed ones are strictly increasing. For discrete data (such as our binary and ordinal variables), the cumulative distribution function is a step function rather than continuous, and the rigorous approach to estimating latent Gaussian correlations would require computationally heavy and numerically unstable methods.

Instead, we adopt an approximation by treating the discrete variables as continuous variables with a massive number of ties. In this implementation, the first step is to estimate the latent Gaussian correlation matrix using a tie-corrected rank correlation. By assigning the average rank to tied values (e.g., all zeros in a binary variable receive the same average rank), the ranking algorithm neutralizes the effects of severe skewness and bypasses the messy marginal scaling of the observed variables.

Let $\hat{\rho}_{uv}^S$ denote this tie-corrected sample Spearman correlation between variables u and v . For a sample of n observations, the estimator is the sample Pearson correlation computed directly on

their average ranks:

$$\hat{\rho}_{uv}^S = \frac{\frac{1}{n} \sum_{i=1}^n \left(\frac{\text{rank}(u_i)}{n+1} - \frac{1}{2} \right) \left(\frac{\text{rank}(v_i)}{n+1} - \frac{1}{2} \right)}{\sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{\text{rank}(u_i)}{n+1} - \frac{1}{2} \right)^2} \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{\text{rank}(v_i)}{n+1} - \frac{1}{2} \right)^2}}, \quad (1)$$

where $\text{rank}(\cdot)$ uses the average method for ties. While in the exact continuous setting (where ties occur with probability zero) this mathematically simplifies to the classical $1 - \frac{6}{n(n^2-1)} \sum d_i^2$ formula, the full tie-corrected formulation is strictly required for our mixed data.

Following the Spearman version of the Harris–Drton construction, the corresponding latent Gaussian correlation is estimated by applying a known trigonometric transformation to the tie-corrected Spearman statistic:

$$\hat{\rho}_{uv} = 2 \sin \left(\frac{\pi}{6} \hat{\rho}_{uv}^S \right). \quad (2)$$

Applying this transformation to every pair of variables yields an estimated latent correlation matrix $\hat{\Sigma}$. Even though the theoretical assumptions are somehow relaxed, it preserves the relative strength of associations and pushes the correlation values in the correct direction. Because constraint-based causal discovery algorithms (like PC and IAMB) rely primarily on the binary outcome of hypothesis tests rather than the exact magnitude of the partial correlations, this approximation is sufficient for recovering the skeleton of the underlying causal graph.

Once $\hat{\Sigma}$ has been constructed, conditional-independence testing proceeds through low-order partial correlations. For any pair of variables (u, v) and conditioning set S , let

$$\hat{\Psi} = \hat{\Sigma}_{(u,v,S),(u,v,S)} \quad (3)$$

be the principal submatrix indexed by u, v , and the variables in S . The estimated partial correlation is then obtained through the precision-matrix identity:

$$\hat{\rho}_{uv|S} = -\frac{\hat{\Psi}_{uv}^{-1}}{\sqrt{\hat{\Psi}_{uu}^{-1} \hat{\Psi}_{vv}^{-1}}}. \quad (4)$$

Under the latent-Gaussian interpretation, testing whether X_u and X_v are conditionally independent given X_S reduces to testing whether the corresponding partial correlation is zero. We therefore use the Fisher- z statistic

$$Z_{uv|S} = \frac{1}{2} \log \left(\frac{1 + \hat{\rho}_{uv|S}}{1 - \hat{\rho}_{uv|S}} \right) \sqrt{n - |S| - 3}, \quad (5)$$

and compare $|Z_{uv|S}|$ with the relevant standard-normal cutoff.

This rank-based construction provides a single conditional-independence engine that can then be used by both the global and the local discovery procedures.

4.2 Causal Discovery Algorithms

The goal of the discovery stage is not to reconstruct a complete causal graph with full certainty, but to extract compact, outcome-centered adjustment candidates that can be compared with the full-covariate and LASSO-based specifications. Two complementary strategies are used for this purpose: a global graph-search procedure based on Rank-PC and a local blanket-search procedure based on IAMB.

These two approaches answer related but distinct questions. The global procedure asks which edges can be removed from a graph by repeated conditional-independence testing, thereby producing a neighborhood structure around the outcome. The local procedure, by contrast, asks which

variables remain directly informative about Y once conditioning is carried out on an incrementally constructed set. In both cases, the key object of interest is the conditional dependence structure around the outcome Y , since that structure is what drives the choice of compact candidate adjustment sets.

In a DAG, the Markov blanket of Y is the smallest set of variables that renders Y conditionally independent of all other nodes [3]. It is composed of the parents of Y , the children of Y , and the parents of those children. From the standpoint of variable selection, this makes the Markov blanket a natural target: it identifies the local environment of the outcome without requiring the analyst to learn the entire graph exactly.

4.2.1 Global Discovery: Rank-PC Screening Around the Outcome

The PC algorithm [4] learns a graph through a sequence of conditional-independence tests. It begins from a dense graph and progressively removes edges whenever a separating set is found. In classical Gaussian implementations, these tests are based on Pearson partial correlations computed directly from the observed variables. Here, instead, the testing step is driven by the rank-based latent-correlation machinery defined above.

Thus, the Rank-PC procedure proceeds by repeatedly asking whether two variables remain conditionally associated once one conditions on subsets of their neighbors, with each test implemented through the latent partial correlation $\hat{\rho}_{uv|S}$ and the associated Fisher- z statistic. If the null of conditional independence is not rejected, the edge is removed and the corresponding separating set is recorded. In this way, the procedure gradually prunes the graph and produces an estimated dependence structure around the outcome.

Rather than interpreting the resulting graph as a fully resolved causal structure (in which every edge represents a definitive cause-and-effect relationship) we utilize the Rank-PC primarily as a global screening tool to identify a compact neighborhood around the outcome Y . Even when the PC algorithm is theoretically well-posed, its output does not uniquely identify a single causal Directed Acyclic Graph (DAG). Instead, because multiple DAGs can imply the exact same conditional independences, the algorithm estimates a Markov equivalence class, typically represented as a partially directed graph.

Because the Harris-Drton framework assumes continuous data where ties do not occur, our dataset’s ordinal variables and skewed distributions require a more cautious interpretation of the output. Given that the covariates depart from the strict continuous assumptions of the nonparametric framework, we avoid treating the output as a definitive causal map. Instead, Rank-PC provides an observational neighborhood estimate a robust, data-driven subset of variables directly adjacent to the outcome Y . By extracting this neighborhood and excluding the intervention node A , we isolate a highly relevant candidate adjustment set.

4.2.2 Local Discovery: The IAMB Algorithm

Whereas Rank-PC works by pruning a graph globally, the Incremental Association Markov Blanket (IAMB) algorithm [5] focuses directly on the local neighborhood of the target variable Y . Its purpose is to estimate a candidate Markov blanket. In practical terms, IAMB is attractive because it avoids the need to learn the full graph when the inferential goal is only to identify a compact adjustment set centered on the outcome.

The local search is built on exactly the same rank-based conditional-independence machinery introduced above. IAMB uses the same latent correlation matrix, the same partial-correlation

calculation, and the same Fisher- z testing rule; what changes is the search logic. Rank-PC is global and edge-oriented, whereas IAMB is local and blanket-oriented.

Algorithmically, IAMB proceeds in two stages. It begins from an empty candidate blanket and enters a forward phase. All variables not yet included are examined one by one, and their residual association with Y is assessed conditional on the current blanket. The variable displaying the strongest statistically significant association with Y is added. This step is then repeated, each time conditioning on the variables already selected, until no remaining variable shows a sufficiently strong conditional association with the outcome.

If a variable remains associated with Y even after conditioning on the current candidate blanket, then the blanket is plausibly still incomplete. However, because this search is greedy, it may temporarily admit variables that only appear important early on, before more relevant variables have entered the blanket. To correct for this possibility, IAMB follows with a backward phase. In this stage, each variable currently retained in the blanket is retested conditional on the others. If a variable becomes conditionally independent of Y once the rest of the blanket is taken into account, it is removed.

Conceptually, IAMB is especially appealing in problems where the inferential target is local rather than global. If the aim is not to reconstruct the entire graph but to identify a small set of variables that are most directly relevant for adjustment around the outcome, then a blanket-search procedure is often more interpretable than a full graph-learning algorithm.

4.3 Outcome-Focused LASSO Penalization

As an alternative to graphical causal discovery, we employ an ℓ_1 -regularized logistic regression (LASSO) [6] to identify a reduced adjustment set consisting solely of strong outcome predictors. By isolating a set of predictors, we prevent variance inflation and potential positivity violations in the subsequent causal estimators.

Because the ℓ_1 penalty is inherently scale-sensitive, we first standardize all candidate covariates. The LASSO estimator then minimizes the negative log-likelihood augmented by a penalty term on the coefficient vector β :

$$(\hat{\beta}_0, \hat{\beta}) = \arg \min_{\beta_0, \beta} \left\{ -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \sum_{j=1}^p |\beta_j| \right\},$$

where $p_i = \mathbb{P}(y_i = 1 \mid x_i) = \frac{1}{1 + \exp(-(\beta_0 + x_i^\top \beta))}$ and β_0 is the unpenalized intercept.

The penalty parameter λ is data-driven and tuned via k -fold cross-validation. The ℓ_1 regularization induces sparsity in the model, forcing the coefficients of conditionally weak or redundant predictors to exactly zero. Applying this procedure to our data, the outcome-only LASSO shrinks most coefficients and robustly isolates a sparse adjustment set comprising three variables: PAY_0, PAY_2, and PAY_6.

4.4 Estimation of the Average Causal Effect

Once the adjustment set L is identified, we estimate the ACE using two distinct methods.

4.4.1 Outcome Regression (G-computation)

The Outcome Regression (or g-formula) approach relies on modeling the conditional mean of the outcome, defined as $m(a, l) = \mathbb{E}[Y \mid A = a, L = l]$. Under the assumptions of consistency and

conditional exchangeability, the expected potential outcomes can be recovered by standardizing over the empirical distribution of the covariates L . We fit a logistic regression model $\hat{m}(A, L; \hat{\theta}) = \Pr(Y = 1 \mid A, L)$. The ACE is then estimated by contrasting the average model-based predictions, where the treatment is artificially set to 1 and 0 for all units:

$$\widehat{\text{ACE}}_{\text{OR}} = \frac{1}{N} \sum_{i=1}^N \hat{m}(1, L_i; \hat{\theta}) - \frac{1}{N} \sum_{i=1}^N \hat{m}(0, L_i; \hat{\theta}).$$

4.4.2 Inverse Probability Weighting (IPW)

IPW creates a pseudo-population in which the treatment assignment is independent of the measured covariates. We first model the propensity score $e(L_i) = \Pr(A_i = 1 \mid L_i)$ using a logistic regression. To ensure numerical stability and strictly enforce the positivity assumption, the estimated scores are bounded in the interval $[0.001, 0.999]$.

To further reduce variance inflation caused by potentially extreme scores, we use stabilized weights:

$$SW_i = \frac{A_i \Pr(A = 1)}{e(L_i)} + \frac{(1 - A_i) \Pr(A = 0)}{1 - e(L_i)},$$

where the marginal probabilities $\Pr(A = a)$ are estimated by their sample proportions. The ACE is then computed using a Hájek-style estimator, evaluating the difference between the weighted empirical means of the outcome for the treated and untreated groups:

$$\widehat{\text{ACE}}_{\text{IPW}} = \frac{\sum_{i=1}^N A_i Y_i SW_i}{\sum_{i=1}^N A_i SW_i} - \frac{\sum_{i=1}^N (1 - A_i) Y_i SW_i}{\sum_{i=1}^N (1 - A_i) SW_i}.$$

For both OR and IPW, uncertainty is quantified using nonparametric bootstrap confidence intervals with $B = 300$ resamples. It is important to note that these bootstrap intervals condition on the chosen adjustment set L , rather than re-running the variable selection and discovery steps inside each bootstrap sample.

5 Empirical Results and Diagnostics

5.1 Structural Learning and Variable Selection

Applying the discovery algorithms outlined in Section 4 to the dataset yields the empirical adjustment sets summarized in Figure 2.

The Rank-PC screening step identifies the direct empirical neighborhood of the outcome as

$$\text{Ne}(Y) = \{A, \text{PAY_0}\}.$$

Removing the treatment node A leaves a highly parsimonious Rank-PC adjustment set consisting solely of the most recent repayment status, PAY_0 .

In contrast, the local IAMB algorithm yields a richer Markov blanket for the outcome:

$$\text{MB}(Y) = \{A, \text{PAY_0}, \text{PAY_2}, \text{BILL_AMT1}\}.$$

After excluding the intervention node A , the corresponding local-discovery adjustment set is

$$L_{\text{IAMB}} = \{\text{PAY_0}, \text{PAY_2}, \text{BILL_AMT1}\}.$$

Thus, the global and local discovery methods do not fully agree in this implementation. While Rank-PC isolates a very compact neighborhood around the outcome, IAMB retains additional variables that remain conditionally associated with Y after its sequential forward and backward search steps. As anticipated, this discrepancy arises from the different conditioning strategies of the two algorithms, but the inclusion of these specific predictors is substantively plausible: PAY_0 captures the most recent and dominant repayment state, whereas PAY_2 and $BILL_AMT1$ may still contain residual information about historical repayment dynamics and financial burden that becomes apparent only when conditioning on the full active set.

Table 1: Local IAMB search trace for the target variable Y . The algorithm keeps adding the most associated variable conditional on the current blanket until no candidate is significant at $\alpha = 0.05$, then performs a backward check.

Phase	Variable / candidate	Partial corr.	p -value
Forward 1	PAY_0	0.354	< 0.001
Forward 2	A	0.134	0.003
Forward 3	PAY_2	0.078	0.081
Backward	PAY_0	0.356	< 0.001
Backward	A	0.134	0.003

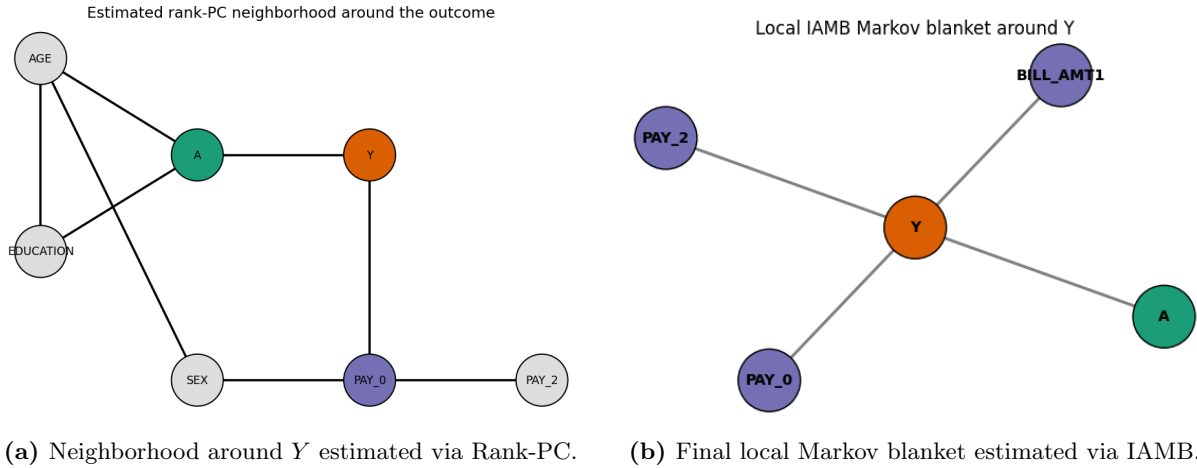


Figure 2: Discovery-based screening outputs around the outcome.

5.2 IPW Diagnostics: Positivity and Covariate Balance

Before interpreting the estimated Average Causal Effect, it is critical to assess the stability of the Inverse Probability Weighting estimator. Figure 3(a) displays the empirical distribution of the estimated propensity scores under the full-covariate model. The estimated probabilities $\hat{e}(L)$ lie roughly within the interval $[0.006, 0.942]$, remaining comfortably within the artificially bounded support of $[0.001, 0.999]$ defined in Section 4.4.2. As a result, the maximum stabilized weight is relatively moderate (approximately 11.65). While there is substantial overlap between the treated and untreated units across most of the support, the distributions thin out in the tails. This confirms that while the structural positivity assumption is not catastrophically violated, the finite-sample IPW estimator can still be sensitive to units with extreme weights.

This sensitivity is directly reflected in the covariate balance diagnostics shown in Figure 3(b). Weighting substantially mitigates the major baseline imbalances, pulling the standardized mean differences (SMDs) for the strongest confounders well below the conventional 0.1 threshold. Most notably, the absolute SMDs for AGE, EDUCATION, and LIMIT_BAL drop precipitously from 0.867, 0.410, and 0.305 to 0.080, 0.000, and 0.009, respectively.

However, balance does not improve uniformly across the high-dimensional covariate space. For instance, the absolute SMD for PAY_AMT6 actually worsens after weighting, increasing from 0.123 to 0.223, and similar mild variance inflation is visible for PAY_AMT4. This diagnostic pattern illustrates the inherent finite-sample limitations of the full-covariate IPW model: forcing balance on a large and potentially noisy set of features can inadvertently inflate imbalances in others, underscoring the classic bias-variance trade-off in propensity score adjustment.

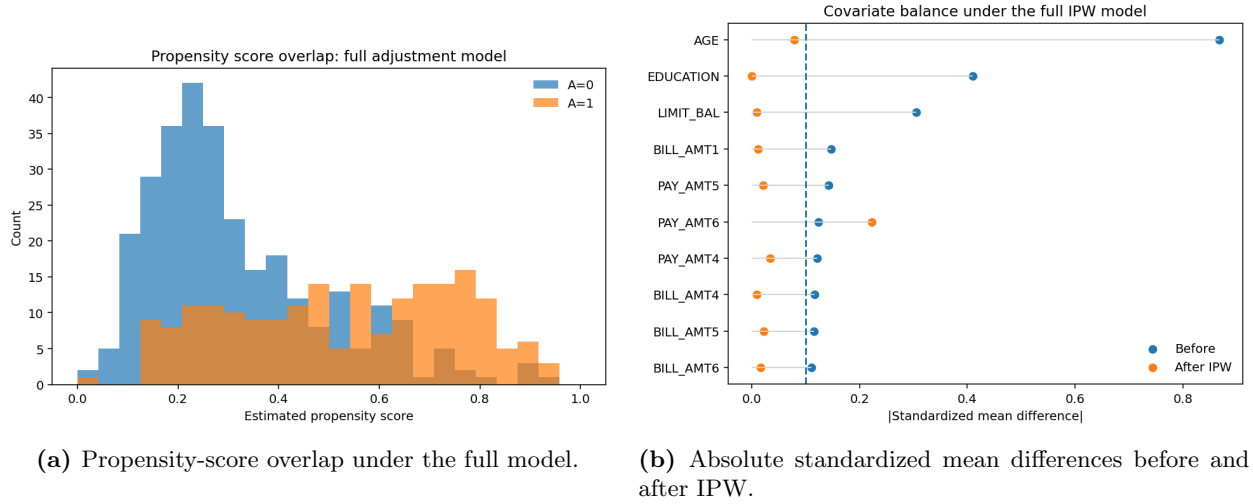


Figure 3: Diagnostics for the full-covariate IPW model.

5.3 Average Causal Effect (ACE) Estimation

The estimated causal effects, alongside their 95% nonparametric bootstrap confidence intervals, are summarized in Table 2 and visually compared in Figure 4. Before adjustment, the crude risk difference stands at +0.1159.

Table 2: Estimated average causal effects and bootstrap confidence intervals using 300 resamples.

Adjustment set	Method	Size	ACE (95% CI)
Crude difference	Unadjusted	0	0.116 (–)
All covariates	Outcome regression	22	0.119 ([0.030, 0.209])
All covariates	IPW	22	0.139 ([0.035, 0.260])
LASSO union	Outcome regression	11	0.118 ([0.038, 0.203])
LASSO union	IPW	11	0.132 ([0.054, 0.229])
PC blanket	Outcome regression	1	0.115 ([0.032, 0.190])
PC blanket	IPW	1	0.118 ([0.041, 0.191])
Local IAMB blanket	Outcome regression	1	0.115 ([0.036, 0.191])
Local IAMB blanket	IPW	1	0.118 ([0.045, 0.192])

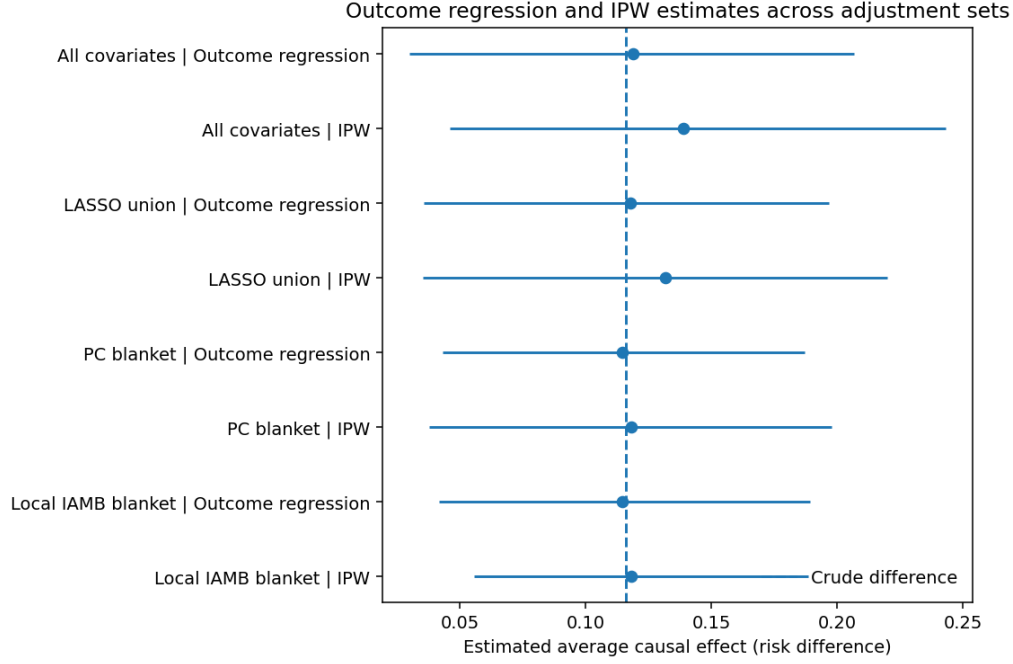


Figure 4

The empirical results reveal three consistent patterns across the causal identification strategies. First, the estimated Average Causal Effect (ACE) remains strictly positive, with the lower bounds of the 95% confidence intervals consistently above zero across all adjustment sets and both estimators. This provides robust statistical evidence supporting the same qualitative conclusion: PPP enrollment is causally associated with an elevated risk of subsequent default, rather than acting as a protective financial measure.

Second, while Outcome Regression (OR) and Inverse Probability Weighting (IPW) yield estimates of similar order of magnitude, the IPW estimates are systematically larger and exhibit wider confidence intervals. This behavior reflects the known finite-sample properties of IPW, where variations in propensity scores (even when stabilized) can inflate the variance of the estimator. Conversely, the OR estimator remains more precise by relying on the stable interpolation of the parametric conditional mean model.

Third, the magnitude of the estimated effect does not simply scale with the dimensionality of the adjustment set, illustrating the complexities of confounding control. The point estimates across the configurations are:

- Full model ($p = 22$): OR = 0.1189, IPW = 0.1388;
- LASSO model ($p = 3$): OR = 0.1162, IPW = 0.1216;
- Rank-PC model ($p = 1$): OR = 0.1145, IPW = 0.1183;
- IAMB model ($p = 3$): OR = 0.1216, IPW = 0.1283.

The full-covariate model produces the largest estimated effects, potentially suffering from mild variance inflation or the inclusion of variables that are strictly outcome predictors rather than true confounders. In contrast, the highly parsimonious Rank-PC neighborhood yields the most conservative estimates. The LASSO and local IAMB models lie between these two extremes,

with the IAMB estimates structurally closer to the full model. Ultimately, while discovery-based variable selection slightly attenuates the estimated effect, the core finding—an approximate 11.5 to 13.8 percentage-point increase in default risk—is structurally robust to the choice of adjustment strategy.

6 Discussion and Conclusion

6.1 Substantive Implications

Across all evaluated models and adjustment strategies, the estimated Average Causal Effect (ACE) of Payment Protection Plan (PPP) enrollment on next-month default risk remains robustly positive, ranging from approximately +0.114 to +0.139. In a strict empirical sense, the analysis provides consistent evidence that PPP enrollment is associated with a higher, rather than lower, short-term default risk.

Economically, the most plausible interpretation of this positive causal estimate is the presence of severe adverse selection. Customers who actively choose to enroll in a PPP are likely those who privately anticipate impending financial distress. Even after adjusting for a rich set of observed historical repayment behaviors and credit limits, the PPP indicator effectively acts as a marker of latent financial fragility rather than a structurally protective intervention. While the exact point estimates fluctuate depending on the adjustment set, the strictly positive confidence intervals decisively rule out the hypothesis that the plan mitigates default risk in practice.

6.2 Methodological Insights: Discovery as a Screening Tool

A core methodological contribution of this project is the integration of rank-based causal discovery to handle the severe non-normality of financial microdata. By leveraging a nonparanormal (Gaussian copula) framework, we bypassed the unstable marginal distributions of skewed monetary variables and ordinal repayment histories.

However, the divergence between the global Rank-PC screening ($\{\text{PAY}_0\}$) and the local IAMB Markov blanket ($\{\text{PAY}_0, \text{PAY}_2, \text{BILL_AMT1}\}$) provides a crucial lesson in applied causal inference. Even when utilizing the exact same underlying latent correlation matrix and partial-correlation testing machinery, global and local search heuristics converge to different sets due to their distinct conditioning strategies. Rank-PC prunes edges aggressively upon finding any valid separating set, while IAMB’s greedy forward-backward search retains variables that maintain residual associations given the active blanket. Consequently, these algorithms should not be viewed as infallible oracles recovering the true underlying Directed Acyclic Graph (DAG), but rather as mathematically principled, data-driven dimension reduction techniques that propose candidate adjustment sets to alleviate over-parameterization.

6.3 Estimator Robustness and Limitations

The empirical contrast between Outcome Regression (OR) and Inverse Probability Weighting (IPW) perfectly illustrates the classic bias-variance trade-off in causal estimation. While OR relies on the parametric extrapolation of the conditional expectation function $\mathbb{E}[Y \mid A, L]$ and yields tighter confidence intervals, IPW attempts to physically balance the covariate space. As shown in the full-covariate model diagnostics, while IPW successfully neutralized major confounders like AGE and EDUCATION, the finite-sample instability of the stabilized weights inadvertently inflated the variance and worsened the standardized mean difference of PAY_AMT6.

Beyond estimator mechanics, the validity of our causal claims rests on strong structural assumptions. Foremost is the assumption of conditional exchangeability ($Y^a \perp\!\!\!\perp A \mid L$). If there are unmeasured confounders—such as sudden job loss, undisclosed external debt, or acute health crises—that simultaneously trigger both PPP enrollment and credit default, all estimated risk differences remain biased. Furthermore, the Harris–Drton rank-PC framework theoretically assumes continuous variables where identical values occur with probability zero. In our dataset, the heavy presence of ties in ordinal variables (e.g., PAY_0 to PAY_6) introduces structural noise into the Spearman correlations, meaning the mapping to the latent Gaussian space operates as a pragmatic approximation rather than an exact mathematical truth.

6.4 Concluding Remarks

This study formally evaluated the causal impact of Payment Protection Plan enrollment on credit card defaults using g-computation and IPW across four distinct adjustment sets (Full, LASSO, Rank-PC, and IAMB). By incorporating a nonparanormal, rank-based discovery step, we addressed the distributional challenges inherent in financial data. Ultimately, all identification strategies converge on the same substantive conclusion: PPP enrollment does not causally reduce default risk but is instead strongly associated with an elevated probability of default, likely driven by unobserved financial distress. While graphical discovery provides a mechanism for confounder screening, this analysis underscores the necessity of interpreting algorithmic outputs through the lens of economic theory, finite-sample diagnostics, and domain expertise.

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